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# Multi-locus genome-wide association studies for root system architectural traits in Ethiopian sorghum (*Sorghum bicolor* L.) landraces

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## Abstract

**Background** Drought is a major constraint to sorghum production. Developing drought-tolerant sorghum varieties is crucial. Optimal root system architecture (RSA) plays a vital role in plant adaptation and productivity under water-limited environment. Genetic assessments of untapped Ethiopian core sorghum germplasm to identify quantitative trait nucleotides (QTN) for RSA traits are crucial. Therefore, this study aimed to identify genetic markers associated with seven RSA and related traits in 182 Ethiopian core sorghum-landraces using multi locus genome-wide association studies (ML-GWAS) using SNP markers.

**Results** The germplasm exhibited substantial phenotypic variation for all measured traits. Genome-wide association analysis identified 181 significant QTNs ( $\text{LOD} \geq 4$ ) associated with nodal root angle (NRA), shoot traits (length, number, dry weight), and RSA (angle, number, dry weight) across six multi-locus GWAS models. Notably, 73 of these QTNs were consistently detected by at least two ML-GWAS methods. A total of 132 potential candidate genes, ranging from 13 to 25 genes per marker trait for the QTNs, were found about 18kbp LD decay of the significant QTNs using biomaRt. Of these, 13 genes were functionally annotated and involved in various biological processes.

**Conclusions** Following the validation of candidate genes for root architecture and related traits, the core collection can be utilized to develop water-efficient sorghum varieties.

**Keywords** RSA, Nodal root angle, QTNs, LOD scores, ML-GWAS, SNP, BiomaRt

## Background

*Sorghum bicolor*, a self-pollinating cereal crop belonging to the *Poaceae* family, is a diploid species with  $2n = 20$  chromosomes and a basic chromosome number of  $X = 10$  [1]. It encompasses 10 intermediate races and five primary races: *Bicolor*, *Caudatum*, *Durra*, *Guinea*, and *Kafir* [2]. In Ethiopia, sorghum ranks third in production and contributes substantially to national grain output [3]. It is a multipurpose food security crop cultivated for food, feed, firewood, house construction, and local beer production [4, 5]. Despite its importance and widespread cultivation, productivity remains low.

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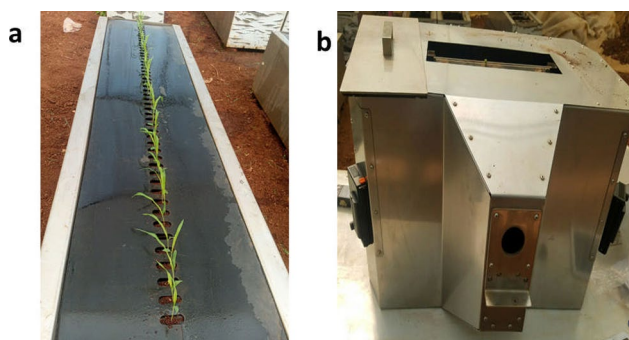
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The major constraints to sorghum production include: diseases, insect pests, striga, moisture stress, and the prevalence of low-yielding local cultivars [6, 7]. Although extensive research has been conducted to improve sorghum production, climate change has exacerbated biotic and abiotic stresses. Among these, drought is the most economically devastating abiotic stress, posing significant challenges to breeders, and its severity is projected to worsen with climate change [8, 9].

Drought is a complex trait with low heritability, making direct yield selection difficult. Consequently, breeders have shifted focus to secondary drought-adaptive traits with moderate to high heritability, which reflect physiological stress responses [10]. These include morphological, physiological, and agronomic traits such as shoot and root architecture, stay-green characteristics, transpiration efficiency, and root system architecture [11–14].

Ethiopia as the center of origin and diversity for sorghum [15], harbors extensive genetic variation across diverse agro-ecologies [16, 17]. Exploring this genetic diversity for drought-related traits in Ethiopian sorghum landraces could facilitate the identification of elite cultivars for developing drought-tolerant varieties [18]. Since drought tolerance involves multiple genes, next-generation sequencing (NGS) technologies have been employed to identify and characterize genes associated with drought resistance, enabling more efficient and targeted breeding [19–23]. While modern techniques have been used to explore core germplasms for above-ground traits like stay-green, limited research has been conducted on the genetic variation and control of root architecture traits for drought tolerance in sorghum [24, 25]. Root system architecture (RSA) refers to the spatial and morphological distribution of plant root systems in soil [25]. An optimal RSA enhances water and nutrient uptake efficiency and plays a critical role in plant adaptation under water-limited conditions [26]. Studies suggest that traits such as fine root diameter, specific root length, root angle, and root length density are key to improve drought resilience [27, 28].



**Fig. 1** Root phenotyping platform (a) root growth chambers (b) high throughput imaging box

The nodal root angle represents a critical component of RSA, determining spatial root distribution within the soil profile [29]. In sorghum, genotypes with narrow nodal root angles and higher root length density at the seedling stage exhibit enhanced drought tolerance [30]. Several studies have identified quantitative trait loci (QTLs) associated with RSA traits, including root length, nodal root angle, root dry weight, root to shoot dry weight ratio and root number [31–35]. Recently, multi-locus GWAS (ML-GWAS) approaches, such as the Multi-Locus Random-SNP-Effect Mixed Linear Model (mrMLM), have emerged as powerful tools for identifying genetic loci linked to complex traits [36]. These methods analyze genome-wide markers while minimizing false positives. However, genetic assessments of Ethiopian core sorghum germplasm for RSA-related QTLs remain limited. Addressing this gap could facilitate the development of water-efficient sorghum varieties tailored to Ethiopian smallholder farmers.

Given the effectiveness of ML-GWAS in dissecting complex traits, this study was initiated to identify genomic regions associated with root angle and related traits and to investigate the relationship between root angle and other drought-related traits.

## Methods

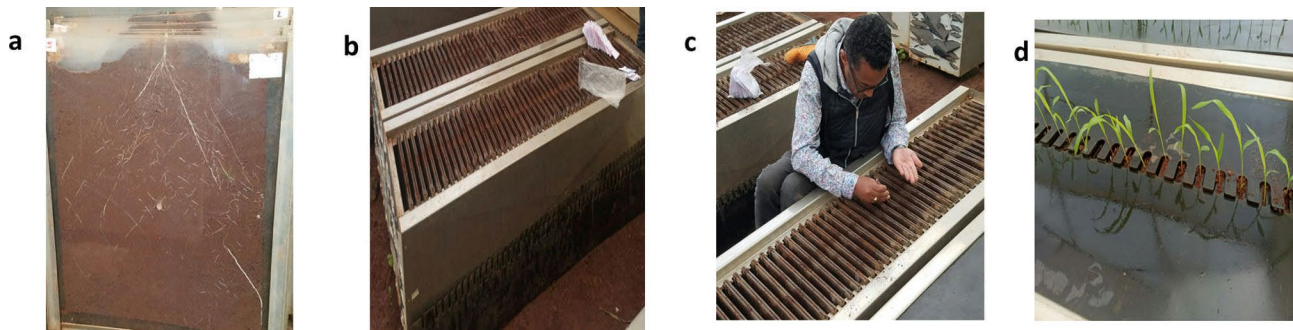
### Plant materials

This study utilized a diverse panel of 190 sorghum accessions sourced from the Ethiopian sorghum germplasm core collection described by [37]. The locations of the accessions and their racial compositions are detailed in Supplementary Table 1. The panel represented a wide spectrum of racial diversity, including the five primary sorghum races: bicolor (8%), caudatum (21%), durra (19%), guinea (17%), and kafir (5%). Additionally, the panel included several intermediate races: namely durra-caudatum (12%), guinea-bicolor (16%), durra-kafir (2%), guinea-caudatum (4%), kafir-bicolor (2%), kafir-caudatum (3%), and durra-guinea (1%) (Table S1).

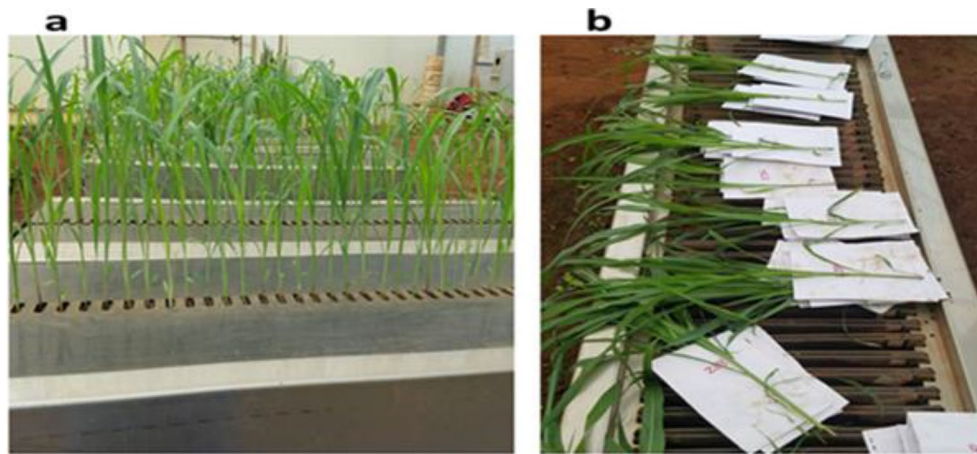
### Growth conditions and experimental design

A root phenotyping experiment utilizing a high-throughput platform was conducted in a white-polycarbonate screen house at the College of Agriculture and Veterinary Medicine, Jimma University, Jimma, Ethiopia in Sep-Nov./2022, Mar-May./2023. The platform is described by [38, 39]. It comprised of 500 root growth chambers connected to 10 metal tubes, with root observation facilitated by black polycarbonate sheets (Fig. 1a). A customized imaging box equipped with dual digital cameras, remotely operated via Android tablets, enabled high-throughput root imaging (Fig. 1b).

Each chamber consisted of two equally sized, parallel transparent sheets (40 cm in height and 35 cm in width),



**Fig. 2** Root phenotyping facility (a) root chamber filled with soil (b) metal tube container to hold root chambers (c) seeds planted in the center of each chamber (d) emergence of seedlings



**Fig. 3** Seedling establishment at screen house (a) six leaves had fully expanded (b) shoot from each chamber was cut at the base of the stem

held in place by a rectangular iron frame with rubber strips. A 3 mm gap between the sheets was maintained and filled with soil (Fig. 2a). Then, the chambers were placed vertically into ten metal tubs each containing 50 chambers (Fig. 2b) and they were watered to its field capacity and three seeds were planted in the center of the chamber by positioning the embryo facing the wall after the seeds were treated with Apron Star 42WS fungicide (Fig. 2c). Next, a black polycarbonate sheet with holes was placed over the tub to block light from reaching the soil and roots, allowing the seedlings to grow and thinned to one after one week. They were watered once or twice a week. Urea and diammonium phosphate (DAP) fertilizer were applied at a rate of 2 gm and 3gm per chamber respectively. The experiment was carried out in partially replicated Row-Column design with two sets of run (160 treatments at once) with the first twenty genotypes replicated four times and the rest 140 were replicated three times.

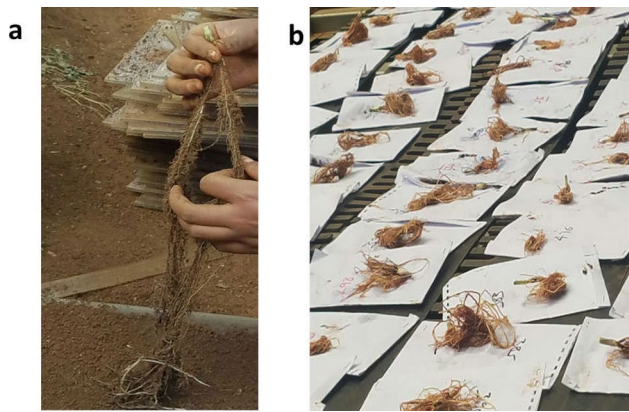
#### Phenotypic characterization

At harvest (six-leaf stage; Fig. 3a), shoots were excised at the stem base (Fig. 3b) for phenotypic measurements. Shoot length (SL) and stem diameter (SD) were recorded

immediately, while shoot dry weight (SDW) was determined after oven-drying at 60 °C for 72 h. Then, each chamber was inserted into camera holding box (Fig. 1b) and root system architecture images were taken on both sides of the chamber using SX600 HS 16MP Ultra-Zoom Digital camera. The angle between the first flush of pair of nodal roots was measured from the images using Opengel photo software, designed by Queensland University [39]. Nodal Root angle (NRA) for a plant was determined from the mean of four observations two from both sides of the chambers. The roots were taken out of the chambers and other types of data like root length (RL) (Fig. 4a), and total number of root (RN) were measured and roots were washed gently and allowed to dry in air (Fig. 4b). Then, root dry weight (RDW) data was taken after drying in oven at 60 °C for 3 days. Moreover, data on root to shoot dry weight ratio (RDW/SDW) was measured by dividing dry root weight to dry shoot weight [27, 30, 35].

#### Phenotypic data analysis

Data were analyzed using a restricted maximum likelihood (REML) mixed model [39]. Predicted values were calculated for each of the eight traits, namely shoot



**Fig. 4** Root architectural traits (a) root length (b) preparation for root oven drying

length (SL), shoot diameter (SD), shoot dry weight (SDW), root length (RL), root number (RN), root dry weight (RDW), ratio of shoot dry weight to root dry weight (RDW/SDW) and nodal root angle (NRA) [40]. The REML mixed model for the traits was carried out using:

$y = X\tau + Zu + e$ ; where  $y$  is the trait of interest,  $\mu$  is the overall mean effect,  $\tau$  is the fixed effect (genotypes),  $X$  is the design matrix for the fixed effect,  $u$  is random effect, and  $Z$  is the design matrix for the random effect and  $e$  is the residual error effects.

The predicted value for each genotype for each trait was performed as fixed effect Best Linear Unbiased Estimators (BLUEs), are statistical estimates that provide the most accurate (minimum variance), linear, and unbiased predictions of fixed effects in a linear model, using META-R [41]. It is also used to compute broad sense heritability ( $H^2$ ) of a given trait at an individual environment is calculated as

$$H^2 = \frac{\sigma_g^2}{(\sigma_g^2 + \sigma_e^2) / nRep}$$

where  $\sigma_g^2$  and  $\sigma_e^2$  are the genotype and error variance components, respectively, and  $nRep$  is the number of replicates.

Phenotypic correlation, the observable correlation between variables, which includes genotypic and environmental effects, was calculated from the variance covariance components using the formula by [42] as follows:

$$\text{Phenotypic correlation} = \frac{\text{phenotypic covariance of } X \text{ and } Y}{\sqrt{(\sigma^2_{pX} \times \sigma^2_{pY})}}$$

where,  $\sigma^2_{ph}$  of  $X$  = phenotypic variance for character  $X$ , and  $\sigma^2_{ph}$  of  $Y$  = Phenotypic variance for character  $Y$ . Linear regression and cluster analyses were carried out using

the R program. Principal component analysis (PCA), linear regression and multiple regressions were performed using the built-in R functions `prcomp` and `princomp`. Moreover, cluster analysis was conducted using the `cluster` R package in the R program (version 4.1.2).

#### Genotyping & quality control

Genotyping by sequencing (GBS) based single nucleotide polymorphism (SNP) markers generated from the sorghum diversity panels and stored in the Purdue University research repository were used for this study [37]. After extracting the SNPs, 546,104 they were subjected to a quality filter following criteria of minor allele frequency (MAF) < 5%, removing SNPs data missing 20% and removing 10% taxa using Tassel software [43]. 70,923 SNPs were used for further analysis.

#### Population structure and kinship analysis

Population structure analysis of 182 sorghum germplasms was performed using 16,383 SNP markers in STRUCTURE v2.3.4 [44]. We employed a model-based clustering approach with  $K$ -values ranging from 1 to 10, running three replicates per  $K$ . Each replicate included a 10,000-step burn-in period followed by 100,000 MCMC iterations. The structure output was imported to Structure Selector to find the optimum  $K$  value for the population [45]. Population admixture was further examined with principal components analysis (PCA) using Plink and plotted the graph with R package SNPRelate ([www.r-project.org](http://www.r-project.org)) [46].

The genetic relatedness between pairs of individuals was measured using a square matrix called a kinship matrix using GAPIT software [47].

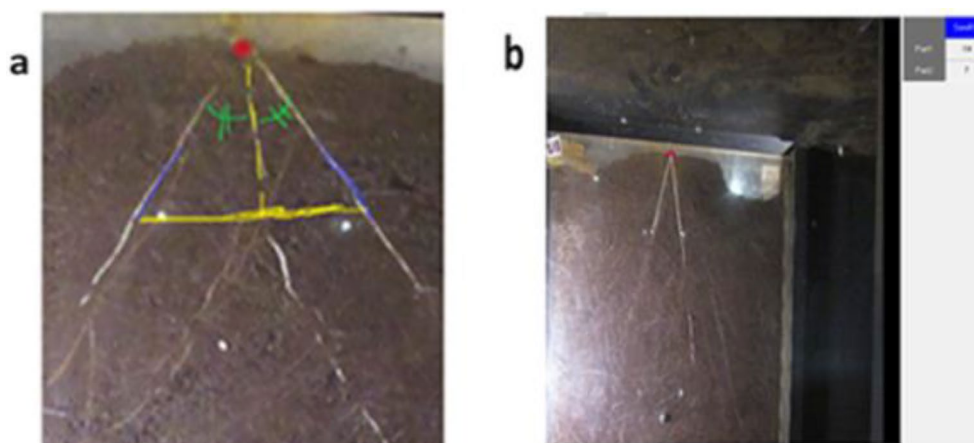
#### Linkage disequilibrium and LD decay analysis

Extent of Linkage disequilibrium (LD) with in core-collection of sorghum germplasms were estimated using squared allele-frequency correlations ( $r^2$ ) value targeting allelic states of two given polymorphic loci using TASSEL v5 software [48]. A sliding window of 50 markers was employed to calculate the correlation between frequencies of SNP pairs. P-values for each  $r^2$  estimate were calculated using 1,000 permutations in the software. The LD decay distance was determined by fitting a nonlinear model, modified by [49] where critical values of  $r^2$  were based on a fixed value of 0.1 and from the parametric 95th percentile of unlinked markers [50]. Using the LOESS curve and a 1 cM moving means, the association between these baseline  $r^2$  values and genetic distance was ascertained. The point at which the LOESS curve first crosses the baseline  $r^2$  value was used to determine genetic distance for the LOESS assessment of LD decay [51]. In addition, heat maps based on  $r^2$  estimates and

**Table 1** Analysis of variance for root angle and related traits

| SOV  | NRA      | SL       | RN      | RL       | RDW      | ShDW    | RDW/ShDW           | ShD     |
|------|----------|----------|---------|----------|----------|---------|--------------------|---------|
| Unit | Degree   | cm       | Count   | cm       | gm       | gm      | gm                 | mm      |
| Df   | 404.0    | 420      | 420     | 420      | 380      | 412     | 384                | 420     |
| Rep  | 612.4    | 740.9    | 42.39   | 566.9    | 273.01   | 10.38   | 0.76               | 44.6    |
| Gen  | 178.7*** | 50.95*** | 5.73*** | 153.8*** | 547.4*** | 1.43*** | 0.69 <sup>ns</sup> | 4.09*** |
| Res  | 112.1    | 11.86    | 2.61    | 65.55    | 271.22   | 0.328   | 0.95               | 0.784   |

\*SOV-source of variation, Df-degree of freedom, Rep-replication, Gen-genotypes, Res- residual, value of mean sum of squared, NRA-nodal root angle, PH-plant height, RN-root number, RL-root length, RDW-root dry weight, ShDW-shoot dry weight, ShD- shoot diameter, RDW/ShDW-ratio of dry weight

**Fig. 5** Representative images showing contrasting root angle: (a) wide-angle (b) narrow-angle architectures

corresponding p values for all marker pairs was plotted across the genome using R program.

#### Genotyping and genome-wide association study

After quality control, a final set of 70,923 highly polymorphic SNP markers and BLUP means of 182 phenotypic data, removing 8 genotypes, were used to conduct GWAS. Six ML-GWAS models namely multi-locus Random-SNP-effect MLM (mrMLM) [52], factored spectrally transformed multi-locus random-SNP-effect MLM (FASTmrMLM) [53], factored spectrally Transformed multi-locus random-SNP effect efficient mixed model association (FAST mr EMMA) [36], polygenic-background-control- based least angle regression plus empirical Bayes (pLARM EB) [54], Iterative Sure Independence Screening EM-Bayesian LASSOISIS (EM-BLASSO) [55], and polygenic-background-control-based Kruskal-Wallis test plus empirical Bayes (pkWmEB) [56] were used for GWAS analysis. The models were implemented using default settings in the R package viz. “multi-locus random-SNP-effect Mixed Linear Model with Graphical User Interface” (mrMLM. GUI) v4.0 program (<https://cran.r-project.org/package=mrMLM>) [57]. Quantitative trait nucleotides (QTNs) with a logarithm of odds (LOD) score of  $\geq 4.00$  (or  $P \leq 0.00002$ ) were regarded as significant to be linked with RA and related traits. In addition, SNP markers which were identified in at least three models were designated as reliable markers [52]. Genomic

positions of QTNs which were identified in this study were compared with previously identified QTNs using sorghum QTL Atlas data base (Sorghum QTL Atlas – OZ Sorghum) within the range of linkage disequilibrium (LD) decay (18 kb). To identify potential candidate genes underlying the significant QTNs associated with seven root related traits, we utilized the Ensembl Plants database using biomaRt R package (<http://bioconductor.org/packages/>) to search for genes located within a 18-kbp window surrounding each significant QTN, with LOD score greater than 4 and identified by more than one mrMLM models. The identified QTLs were graphically presented using Map chart 2.32 software [58]. Moreover, intermediate and final results after the run of the analysis were used to draw Manhattan and QQ plots for the traits.

## Results

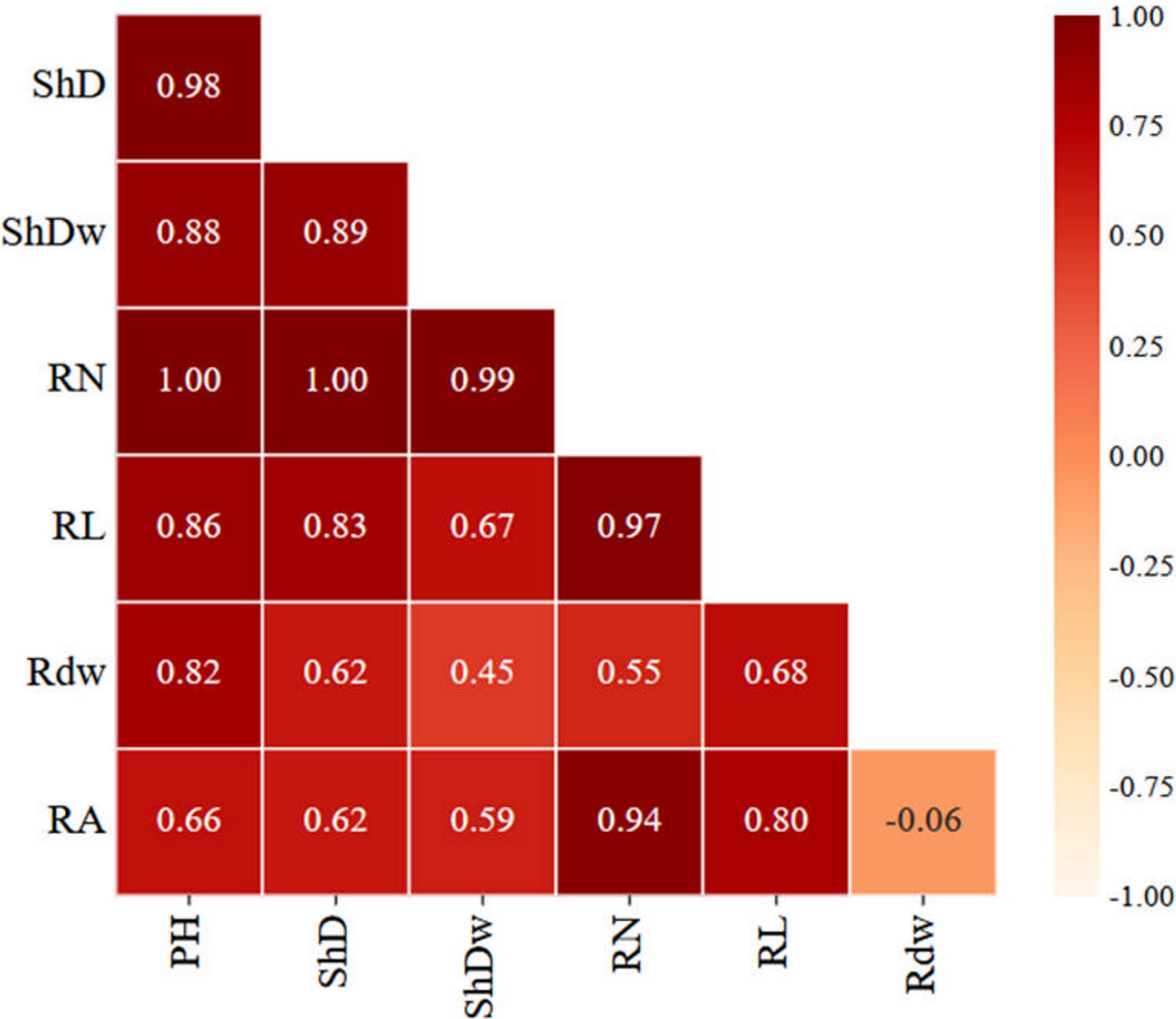
### Phenotypic variation

Analysis of variance (ANOVA) revealed highly significant ( $p < 0.001$ ) genotypic variation for all the traits except root to shoot ratio (Table 1). A wide range of phenotypic variation was observed across the germplasm for the traits. For example, nodal root angle (NRA) exhibited a mean of 44 degree with a range from  $21^{\circ}$  to  $67^{\circ}$  (Fig. 5). Shoot length (SL) ranged from 5 cm to 33 cm, root number (RN) from 2 to 14, root length (RL) from 6 cm to 92 cm, root dry weight (RDW) from 0 g to 4.2 g (Table 2).

**Table 2** Phenotypic variations of traits from BLUP

| Trait | Unit   | Min | Mean  | Max  | G-Var. | Err Var. | H <sup>2</sup> |
|-------|--------|-----|-------|------|--------|----------|----------------|
| NRA   | Degree | 21  | 44    | 67   | 18     | 115.8    | 0.38           |
| SL    | Cm     | 5   | 16.97 | 32.5 | 14.72  | 12.24    | 0.83           |
| RN    | Count  | 2   | 6.68  | 14   | 0.91   | 2.76     | 0.57           |
| RL    | Cm     | 6   | 59.7  | 92   | 27.0   | 68       | 0.6            |
| RDW   | Gm     | 0.1 | 1.5   | 4.2  | 40.9   | 326.5    | 0.33           |
| ShDW  | Gm     | 0.1 | 0.95  | 14.4 | 0.45   | 0.34     | 0.84           |
| ShD   | Gm     | 0.5 | 3.09  | 8.2  | 1.3    | 0.81     | 0.87           |

N.B. Nodal Root angle (NRA), Shoot length (SL), Root number (RN), Root length (RL), Root dry weight (RDW), Shoot dry weight (ShDW), Min (minimum), Max (maximum), G-Var (Genetic variance), Err Var (Residual variance), H<sup>2</sup> (heritability)



**Fig. 6** Genetic correlations of root angle and related traits

**Genetic correlation of nodal root angle and related traits**  
Correlation analysis revealed a strong positive association between root angle and root number (0.94), root length (0.80), Shoot length (0.66), shoot diameter (0.62), and shoot dry weight (0.59) (Fig. 6). Root dry weight

exhibited strong positive correlations with plant height (0.82), root length (0.68), shoot diameter (0.62), root number (0.55), and shoot dry weight (0.45). Moreover, root length showed strong positive correlations with RN (0.97), SL (0.86), ShD (0.83) and ShDW (0.67). Conversely,

RA displayed very weak negative relationships with R<sub>dw</sub> (−0.06).

### Marker distribution across chromosome

After filtering, a total of 70,923 high-quality SNPs were identified across the 10 chromosomes of sorghum. The distribution of these SNPs across chromosomes is presented in Fig. 7 and Supplementary Table S2. The number of SNPs varied across chromosomes, ranging from 5,001 on chromosome 7 to 9,454 on chromosome 1.

### Population structure and kinship analysis

The STRUCTURE result revealed five distinct groups (Fig. 8b). Cluster IV comprised genotypes with both wide (35–35.7°) and narrow (22.8°–26°) root angles, including accessions such as ETSL\_101491 (35.7°), ETSL\_101446 (22.8°), and IS\_38331 (35.4°). Notably, narrow-angle genotypes predominantly belonged to the Durra and Caudatum races, while transpiration efficiency (TE) characterization [59] revealed that Cluster IV contained both high- and low-TE phenotypes. Approximately 30% of accessions exhibited admixture ( $Q < 0.7$ ) across clusters. Likewise, to further investigate population structure,

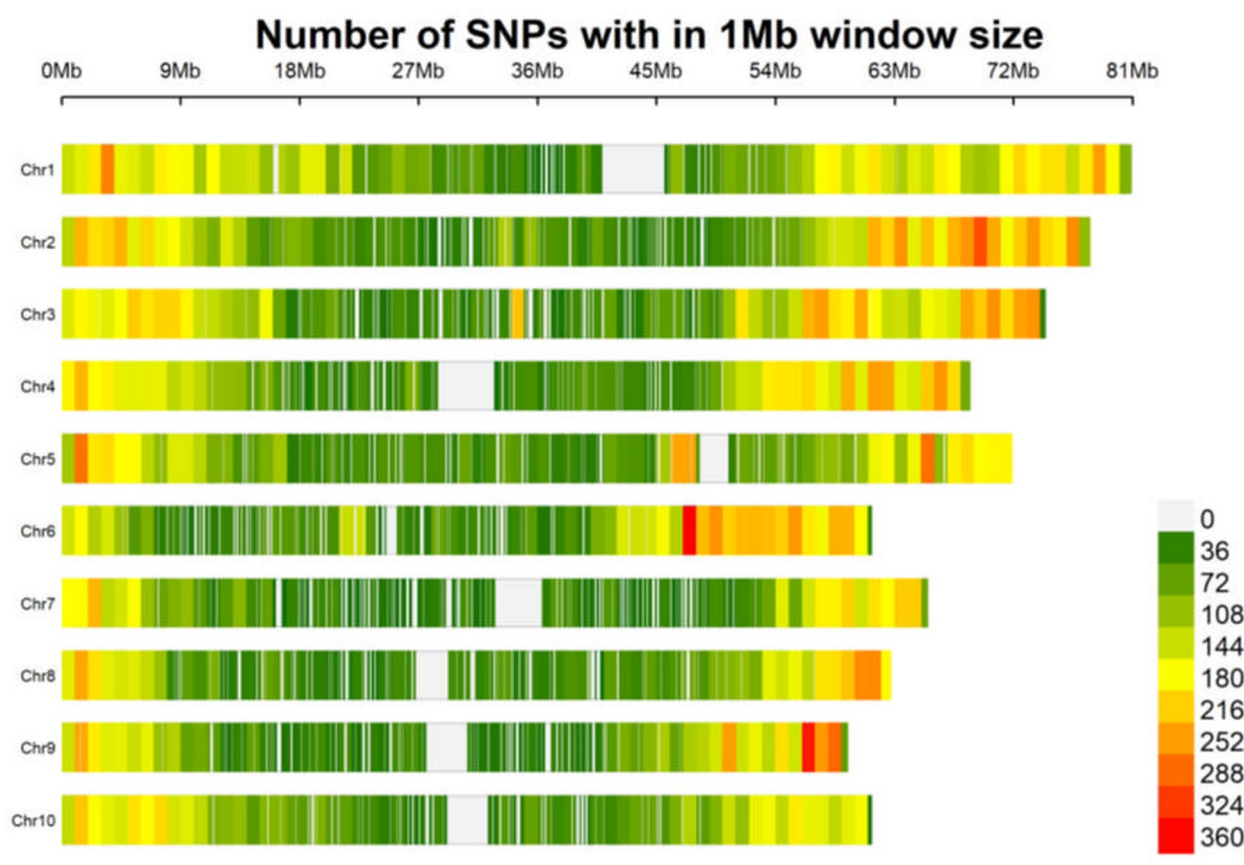
principal component analysis (PCA) was performed and the result showed three groups. The first two principal components (PCs) explained 51% of the total variance, with PC1 and PC2 accounting for 35% and 16%, respectively (Fig. 8c). Q-matrix was used as a structural matrix for the association study. Genetic relatedness among individuals was assessed using a kinship matrix (Fig. 8d). The pairwise kinship coefficients indicated a relatively low level of genetic relatedness among the accessions.

### Linkage disequilibrium decay analysis

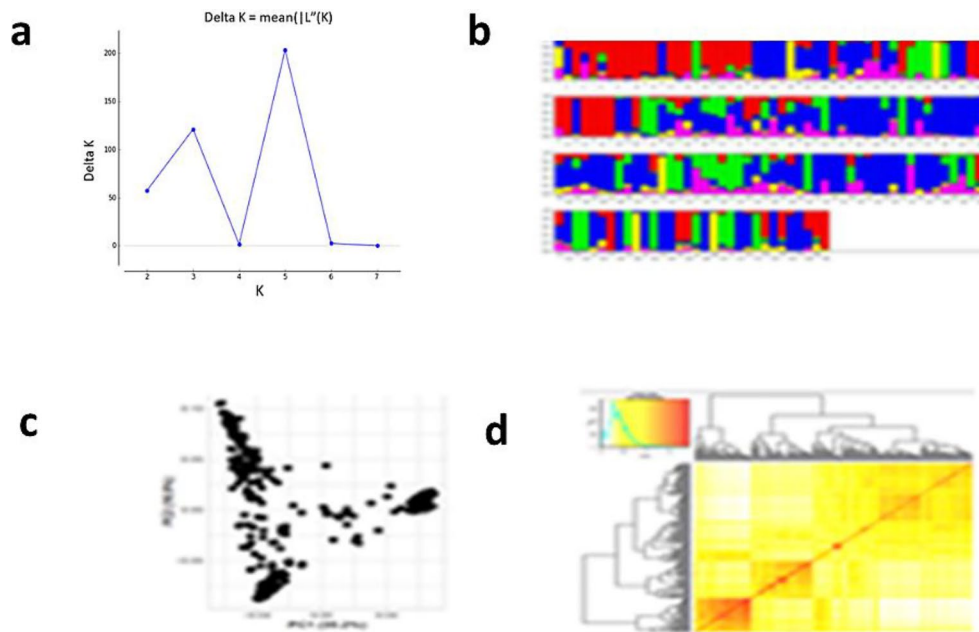
Pairwise linkage disequilibrium (LD) between 70,923 high-quality SNPs was assessed. The genome-wide LD decay rate was estimated 18,301 bp at an  $r^2 = 0.1$  (Fig. 9).

### GWAS analyses

GWAS analysis revealed 181 significant quantitative trait nucleotides (QTNs) with LOD scores greater than or equal to four, associated with seven traits: SL, ShD, ShDw, RN, RL, R<sub>dw</sub>, and RA (Supplementary Table S3). These QTNs were detected using at least one of the six multi-locus genome-wide association study (ML-GWAS)



**Fig. 7** The distribution of filtered SNPs across the 10 chromosomes of sorghum landraces, with colors representing the number of SNPs within each 1 Mb interval



**Fig. 8** Genetic structure and kinship of 182 sorghum landraces **(a)** Optimum number of K groups from Structure Selector,  $K=5$ ; **(b)** Population structure analysis plot for  $K=5$ ; **(c)** PCA of the sorghum landraces; **(d)** Kinship plot of sorghum landraces

methods employed. Q-Q plots were generated to assess the significance of the observed p-values. For most traits, the observed p-values closely followed the expected diagonal line, indicating that the employed GWAS models were generally suitable (Supplementary Fig. 1). QTNs identified per trait ranged from 18 for RN to 37 for RL. Among the ML-GWAS methods employed, pLARM EB identified the highest number of QTLs (72), followed by mrMLM (66), FASTmrMLM (59), ISIS EM-BLASSO (54), pkWmEB (51), and FASTmrEMMA (8) (Supplementary Table S3). A total of 73 significant QTNs were identified by at least two ML-GWAS methods across the seven traits and presented in Manhattan plot (Table 3) (Fig. 10). These QTNs were distributed as follows: eleven for RA, RN, & ShD, thirteen for SL, nine for Rdw, eight for ShDw, and ten for RL. Chromosome 3 harbored the most QTNs (13 QTNs), followed by chromosome 4 (11 QTNs), chromosome 5 (10 QTNs), chromosomes 7 (7 QTNs) and chromosome 1 (6QTNs).

#### Comparison of QTNs from this study and previously identified QTNs

The genomic positions of QTNs which were identified in this study were compared with previously identified QTNs for root related traits viz. RA, Rdw, RL and RN using sorghum QTL Atlas data base (Sorghum QTL Atlas – OZ Sorghum). The result indicated that from 41 QTNs identified, 32 were collocated with previously reported QTLs, while the remaining 9 QTNs were novel (S02\_33207061, S03\_50387459, S04\_39817417, S09\_815477 for root angle, S03\_46797454,

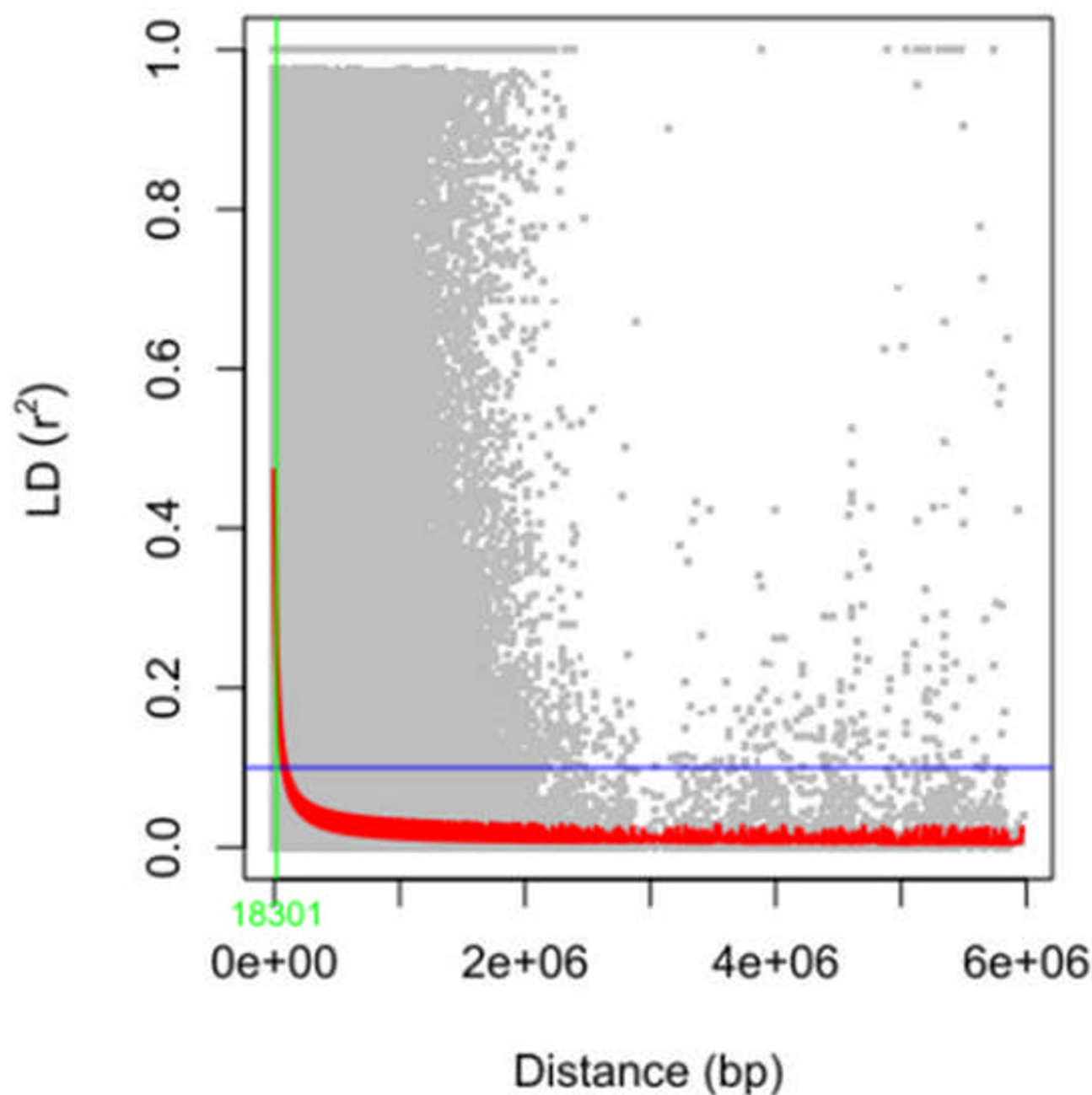
S03\_70595242, S03\_7550823, S05\_1365704 for Rdw, S03\_62263536 for RL) (Tables 3 and 4).

#### Identification of prospective candidate genes associated with root related traits

This analysis was performed using biomaRt package, and successfully identified 132 candidate genes, ranging from 13 to 25 per trait (Supplementary Table S4). Of these, 13 genes were functionally annotated and involved in various biological processes. For example: for the trait RA, two candidate genes (*SORBI\_3009G008200*, *ENSRNA049995717*) and (*SORBI\_3009G251600*, *ENSRNA049443719*) were identified on chromosome 9 (Fig. 11). The first gene is involved in regulations of probable Serine/threonine-protein Kinase while, *SORBI\_3009G251600* is involved in cytochrome coxidase, subunit Vb family protein. For RDW three candidate genes were identified viz. *SORBI\_3010G247800*, *SORBI\_3003G087300*, *SORBI\_3003G236900*, on chromosomes 3, and 10 having different functions. For instance, *SORBI\_3003G236900* plays role in Flavin mono-xxygenase-like enzyme, Auxin biosynthesis. For RL two candidate genes were identified and involved in different functions. For example, *SORBI\_3004G012900* (*ENSRNA049444525*) gene has a role in regulation of protein in low photosystem II accumulation.

#### Discussion

Drought is becoming a major sorghum production constraint in the face of changing climate [66]. Developing drought-tolerant sorghum varieties offers several



**Fig. 9** LD decay analysis of 182 sorghum accessions

significant benefits for food security and sustainable agriculture [67]. Ethiopia is recognized as a primary center of origin and diversity for sorghum [15]. The country's varied agro-ecologies create a unique opportunity for sorghum germplasm to exhibit diverse genetic backgrounds, encompassing nearly all known races of sorghum [16].

This diversity can enhance drought tolerance by exploiting the natural genetic variation found in Ethiopian sorghum germplasms. Consequently, exploring the genetic diversity of Ethiopian sorghum landraces for traits related to drought could facilitate the development

of sorghum varieties that are better suited to moisture stress, ensuring productivity even in conditions of limited water availability [18]. Fortunately, advances in high-throughput genotyping enabled effective harnessing of the genetic diversity of the germplasm [68]. For instance, GWAS is a systematic approach to link genetic variation with complex traits, providing valuable insights that can be used for crop improvement and understand the genetic basis of traits like drought tolerance.

The use of genetic markers has enabled researchers and breeders to improve crop resilience against

**Table 3** Significant QTNs for seven RSA & related traits detected at once by using two or more multi-locus GWAS models

| Trait | QTN          | Chr. | Marker position (bp) | LOD score | R <sup>2</sup> (%) | Method                                                |
|-------|--------------|------|----------------------|-----------|--------------------|-------------------------------------------------------|
| SL    | S01_70014967 | 1    | 70,014,967           | 4         | 3.9                | FASTmrMLM, mrMLM                                      |
|       | S02_44514649 | 2    | 44,514,649           | 5         | 6.3                | ISISEMBLASSO, pKWmeB, pLARmeB                         |
|       | S03_5098770  | 3    | 5,098,770            | 5         | 4.3                | FASTmrMLM, ISIS EM-BLASSO, mrMLM                      |
|       | S03_5547943  | 3    | 5,547,943            | 4         | 5                  | FASTmrEMMA, FASTmrMLM, ISISEM-BLASSO, mrMLM, pLARmeB  |
|       | S04_22455254 | 4    | 22,455,254           | 5         | 6                  | FASTmrMLM, mrMLM                                      |
|       | S05_38395351 | 5    | 38,395,351           | 6         | 9.2                | FASTmrMLM, ISIS EM-BLASSO, pKWmeB,                    |
|       | S05_61217883 | 5    | 61,217,883           | 4         | 3.5                | FASTmrMLM, mrMLM                                      |
|       | S05_70243460 | 5    | 70,243,460           | 5         | 16.5               | pKWmeB, pLARmeB                                       |
|       | S06_48896408 | 6    | 48,896,408           | 5         | 3.3                | FASTmrMLM, ISIS EM-BLASSO, pLARmeB                    |
|       | S06_60571800 | 6    | 60,571,800           | 5         | 2.2                | ISIS EM-BLASSO, pLARmeB                               |
|       | S07_45505726 | 7    | 45,505,726           | 5         | 13                 | FASTmrMLM, ISIS EM-BLASSO, pLARmeB, mrMLM             |
| RA    | S07_58136571 | 7    | 58,136,571           | 8         | 6.9                | FASTmrMLM, mrMLM                                      |
|       | S09_52837691 | 9    | 52,837,691           | 4         | 3.1                | FASTmrMLM, ISIS EM-BLASSO                             |
|       | S01_32326466 | 1    | 32,326,466           | 7         | 9.2                | ISIS EM-BLASSO, pKWmeB                                |
|       | S01_72392128 | 1    | 72,392,128           | 7         | 9.2                | FASTmrMLM, ISIS EM-BLASSO, mrMLM, pKWmeB, pLARmeB     |
|       | S02_33207061 | 2    | 33,207,061           | 4         | 5                  | FASTmrMLM, ISIS EM-BLASSO, mrMLM                      |
|       | S03_50387459 | 3    | 50,387,459           | 9         | 5                  | ISIS EM-BLASSO, pKWmeB                                |
|       | S03_8641122  | 3    | 8,641,122            | 8         | 5                  | ISIS EM-BLASSO, pKWmeB                                |
|       | S04_39817417 | 4    | 39,817,417           | 6         | 7                  | ISIS EM-BLASSO, mrMLM, pKWmeB                         |
|       | S05_43703370 | 5    | 43,703,370           | 6         | 10                 | FASTmrMLM, pLARmeB                                    |
|       | S07_29563576 | 7    | 29,563,576           | 4         | 3                  | FASTmrMLM, mrMLM                                      |
|       | S07_37035244 | 7    | 37,035,244           | 4         | 6                  | FASTmrMLM, ISIS EM-BLASSO, pKWmeB, pLARmeB            |
| Rdw   | S09_58728450 | 9    | 58,728,450           | 4         | 3                  | pKWmeB, pLARmeB                                       |
|       | S09_815477   | 9    | 815,477              | 4         | 4                  | FASTmrEMMA, FASTmrMLM, pKWmeB, pLARmeB                |
|       | S03_46797454 | 3    | 46,797,454           | 4         | 3                  | FASTmrMLM, mrMLM                                      |
|       | S03_57609059 | 3    | 57,609,059           | 8         | 6                  | FASTmrEMMA, FASTmrMLM, ISIS EM-BLASSO, mrMLM, pLARmeB |
|       | S03_70595242 | 3    | 70,595,242           | 13        | 1                  | FASTmrMLM, pLARmeB                                    |
|       | S03_7550823  | 3    | 7,550,823            | 18        | 19                 | FASTmrMLM, pLARmeB                                    |
|       | S04_50974937 | 4    | 50,974,937           | 11        | 12                 | mrMLM, pKWmeB                                         |
|       | S05_1365704  | 5    | 1,365,704            | 5         | 0.3                | mrMLM, pLARmeB                                        |
|       | S05_34473660 | 5    | 34,473,660           | 7         | 0.4                | FASTmrMLM, pLARmeB                                    |
|       | S06_42051530 | 6    | 42,051,530           | 12        | 0.5                | FASTmrMLM, mrMLM, pLARmeB, FASTmrMLM, pLARmeB         |
|       | S10_58800380 | 10   | 58,800,380           | 7         | 3                  | FASTmrMLM, ISIS EM-BLASSO, mrMLM, pKWmeB, pLARmeB     |
| RL    | S01_24721423 | 1    | 24,721,423           | 4         | 6                  | FASTmrEMMA, pKWmeB, pLARmeB                           |
|       | S02_6837132  | 2    | 6,837,132            | 4         | 10                 | mrMLM, pKWmeB, pLARmeB                                |
|       | S03_18979756 | 3    | 18,979,756           | 4         | 4                  | FASTmrMLM, pLARmeB                                    |
|       | S03_62263536 | 3    | 62,263,536           | 4         | 7                  | FASTmrMLM, ISIS EM-BLASSO                             |
|       | S04_1010651  | 4    | 1,010,651            | 4         | 3                  | ISIS EM-BLASSO, pLARmeB                               |
|       | S07_28520448 | 7    | 28,520,448           | 7         | 21                 | pKWmeB, pLARmeB                                       |
|       | S08_4855598  | 8    | 4,855,598            | 4         | 1.5                | FASTmrMLM, pLARmeB                                    |
|       | S09_57188887 | 9    | 57,188,887           | 8         | 5                  | mrMLM, pLARmeB                                        |
|       | S10_16866728 | 10   | 16,866,728           | 5         | 2                  | ISIS EM-BLASSO, pLARmeB                               |
|       | S10_56475869 | 10   | 56,475,869           | 9         | 20                 | FASTmrMLM, pKWmeB, pLARmeB                            |
|       | S01_5984592  | 1    | 5,984,592            | 5         | 4                  | ISIS EM-BLASSO, mrMLM, pKWmeB                         |
| RN    | S01_72391791 | 1    | 72,391,791           | 4         | 8                  | pKWmeB, pLARmeB                                       |
|       | S03_63149699 | 3    | 63,149,699           | 5         | 3                  | FASTmrMLM, ISIS EM-BLASSO, pLARmeB                    |
|       | S04_56437057 | 4    | 56,437,057           | 5         | 8                  | FASTmrEMMA, FASTmrMLM, pLARmeB                        |
|       | S05_16129064 | 5    | 16,129,064           | 6         | 6                  | FASTmrMLM, pKWmeB, pLARmeB                            |
|       | S05_3566674  | 5    | 3,566,674            | 6         | 5                  | FASTmrMLM, ISIS EM-BLASSO, pKWmeB                     |
|       | S05_45567244 | 5    | 45,567,244           | 5         | 12                 | FASTmrMLM, mrMLM, pLARmeB                             |
|       | S05_67059095 | 5    | 67,059,095           | 6         | 7                  | FASTmrMLM, ISIS EM-BLASSO, pKWmeB, pLARmeB            |
|       | S07_58136512 | 7    | 58,136,512           | 4         | 1                  | FASTmrMLM, ISIS EM-BLASSO, pLARmeB                    |

**Table 3** (continued)

| Trait | QTN          | Chr. | Marker position (bp) | LOD score | R <sup>2</sup> (%) | Method                                      |
|-------|--------------|------|----------------------|-----------|--------------------|---------------------------------------------|
| ShD   | S10_16866728 | 10   | 16,866,728           | 4         | 1                  | ISIS EM-BLASSO, pLARmEB                     |
|       | S10_22306651 | 10   | 22,306,651           | 6         | 5                  | FASTmrMLM, ISIS EM-BLASSO, pKWmEB, pLARmEB  |
|       | S02_48843211 | 2    | 48,843,211           | 6         | 11                 | FASTmrMLM, pKWmEB, pLARmEB                  |
|       | S02_6837132  | 2    | 6,837,132            | 6         | 13                 | pKWmEB, pLARmEB                             |
|       | S03_5547943  | 3    | 5,547,943            | 8         | 6                  | FASTmrEMMA, ISIS EM-BLASSO, pKWmEB, pLARmEB |
|       | S03_63144057 | 3    | 63,144,057           | 5         | 4                  | ISIS EM-BLASSO, pKWmEB                      |
|       | S04_13852192 | 4    | 13,852,192           | 4         | 2                  | FASTmrMLM, mrMLM, pKWmEB                    |
|       | S04_44711266 | 4    | 44,711,266           | 6         | 13                 | FASTmrMLM, mrMLM, pLARmEB                   |
|       | S06_48896408 | 6    | 48,896,408           | 7         | 5                  | FASTmrMLM, ISIS EM-BLASSO                   |
|       | S07_4264143  | 7    | 4,264,143            | 4         | 4                  | FASTmrMLM, ISIS EM-BLASSO, pKWmEB, pLARmEB  |
| ShDw  | S07_58136571 | 7    | 58,136,571           | 17        | 15                 | FASTmrMLM, ISIS EM-BLASSO, pKWmEB, pLARmEB  |
|       | S09_50178715 | 9    | 50,178,715           | 6         | 4                  | FASTmrMLM, pKWmEB, pLARmEB                  |
|       | S10_50558691 | 10   | 50,558,691           | 5         | 4                  | FASTmrMLM, ISIS EM-BLASSO, pLARmEB          |
|       | S02_48843211 | 2    | 48,843,211           | 7         | 11                 | FASTmrMLM, mrMLM, pKWmEB, pLARmEB           |
|       | S03_55291405 | 3    | 55,291,405           | 6         | 7                  | mrMLM, pLARmEB                              |
|       | S03_70199859 | 4    | 70,199,859           | 4         | 1                  | FASTmrMLM, mrMLM                            |
|       | S04_19691402 | 4    | 19,691,402           | 9         | 24                 | FASTmrMLM, ISIS EM-BLASSO, mrMLM, pLARmEB   |
|       | S04_22246701 | 4    | 22,246,701           | 6         | 8                  | ISIS EM-BLASSO, mrMLM, pKWmEB               |
|       | S04_42656485 | 4    | 42,656,485           | 4         | 5                  | FASTmrMLM, mrMLM                            |
|       | S07_4264143  | 7    | 4,264,143            | 6         | 9                  | ISIS EM-BLASSO, pKWmEB, pLARmEB             |
|       | S09_50178715 | 9    | 50,178,715           | 6         | 1                  | pKWmEB, pLARmEB                             |

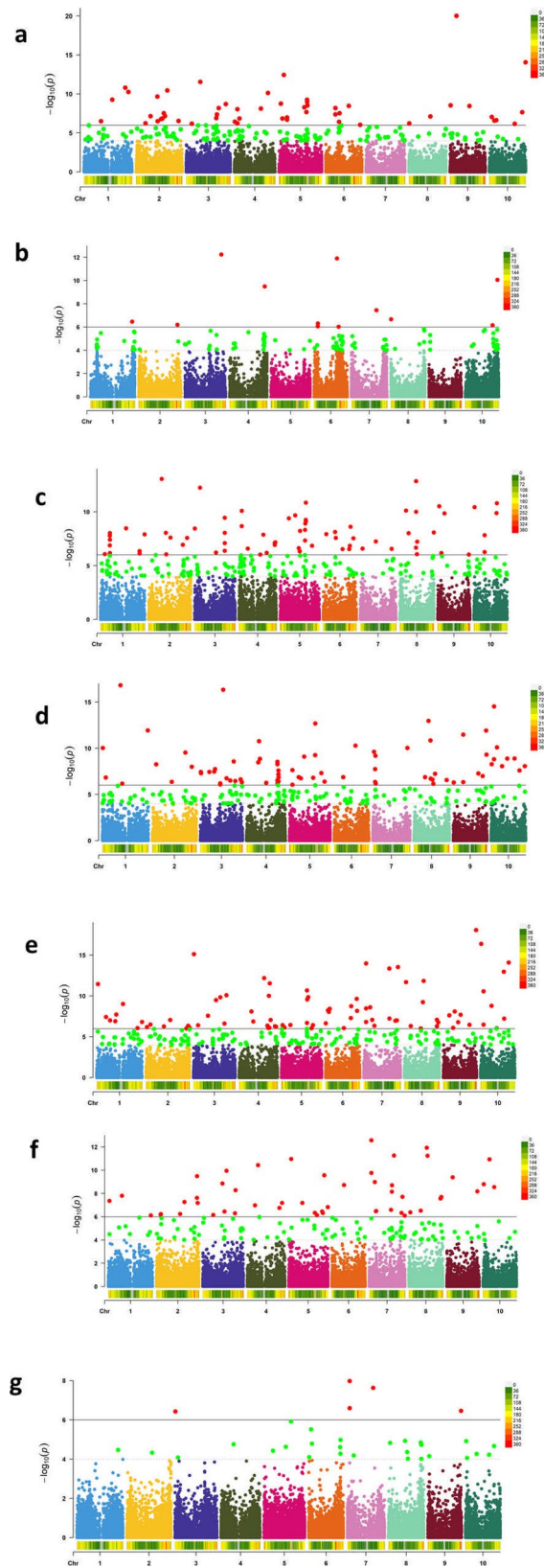
climate-related stresses [69]. Multi-locus Genome-Wide Association Studies (ML-GWAS) now provide significant advantages over conventional single-marker approaches for identifying genetic associations with complex traits. Unlike single-locus methods, ML-GWAS can simultaneously analyze interactions between multiple genetic variants and a critical capability for reading polygenic trait architectures. This approach offers greater statistical power to detect associations, particularly for traits controlled by numerous small-effect alleles, while reducing false positives through advanced modeling. By capturing cumulative genetic effects across the genome, it enables more reliable identification of true associations and provides a more comprehensive understanding of the genetic basis of agriculturally important traits. These capabilities make ML-GWAS an indispensable tool for modern crop improvement programs facing climate change challenges [36, 70].

Our investigation reveals significant genetic variation in root architecture traits among Ethiopian sorghum landraces, particularly in nodal root angle phenotypes (Table 1). The identification of both wide- and narrow-root angled accessions provides critical genetic material for drought tolerance breeding programs, verified by earlier findings [71, 72]. Nodal root angle serves as a crucial drought adaptation mechanism by determining soil exploration strategies for water and nutrient acquisition [73]. Moreover, according to [74], heritability of most of the traits like RA, RN, RL and RDW are moderate; while

PH, ShD and ShDW showed high heritability; root to shoot dry weight showed very low heritability (Table 2).

Root angle variation creates distinct water acquisition strategies that are ecologically adaptive. Shallow, wide-angle root systems excel in environments with intermittent rainfall by efficiently capturing surface moisture [30], providing crucial early-growth drought tolerance. These architectures prove particularly valuable in regions with unreliable precipitation. In contrast, steep, narrow-angle root systems penetrate deeper soil layers to access residual moisture during extended droughts [39], ensuring grain development in arid conditions. This fundamental deal in root architecture underscores the need for environment-specific breeding approaches to enhance drought resilience.

Correlation analysis revealed strong positive associations between nodal root angle and both root number ( $r=0.94$ ) and root length ( $r=0.80$ ) (Fig. 6). These findings align with previous reports documenting similar relationships in sorghum root architecture [24, 75]. According to [76], there was a genetic linkage between root angle other root traits, QTLs co-regulating root angle, number, and length in legumes. Moreover, root angle and plant height showed high correlation implies it is possible to improve the traits at the same time as some genes for height may influence root angle [77]. He explained that QTLs for root angles were co-localize with plant height implies pleiotropy impacting drought adaptation. In addition, high positive correlation was observed between root



**Fig. 10** Genome-wide association analyses of RSA traits using multi-locus methods. Manhattan plots for: (a) RA, (b) Rdw, (c) RL, (d) RN, (e) ShD, (f) ShDw, and (g) SL. Results combine three multi-locus approaches: mrMLM, FASTmrMLM, and FASTmrEMMA. The horizontal dashed line indicates the genome-wide significant threshold. Highlighted loci (red dots) denote statistically significant associations. Points represent median  $-\log_{10}(p)$  values across methods, each point represents a genetic variant, with chromosomes arranged along the x-axis

length, with root number (0.97), plant height (0.86), and shoot diameter (0.83) and shoot dry weight (0.67).

This positive correlation result among the variables implies that these variables are interconnected to each other's in plant growth. Understanding these relationships allows for better insights into plant development, water usage, and productivity [72]. Linkage disequilibrium (LD) decay describes the rate at which LD decreases with increasing genetic distance. Understanding LD patterns is essential for genome-wide association studies (GWAS), and quantitative trait locus (QTL) mapping, particularly for complex traits like root system architecture (RSA) [24].

Genome-wide LD analysis revealed an LD decay distance of 18,301 bp at  $r^2 = 0.1$  in this study, indicating relatively rapid LD decay characteristic of our diverse landrace collection (Fig. 9). For candidate gene identification within significant QTN regions, we applied 18 kb, unlike previous study for the same population a 66 kb LD window by [37]. These findings align with known patterns of LD variation in sorghum: rapid LD decay (~10 kb in landraces; ~100 kb in improved lines) reflecting population structure differences [71]; consistently faster decay observed in wild accessions and landraces compared to elite inbred lines [78].

The key advantage of multi-locus genome-wide association studies (ML-GWAS) is its ability to identify complex genetic architectures underlying traits [79]. This is especially important for traits influenced by multiple genes, each with small effect [70]. In this study, 73 QTNs were identified in more than one ML-GWAS methods for the seven traits (Table 3). These QTNs were distributed mostly all over the sorghum chromosomes.

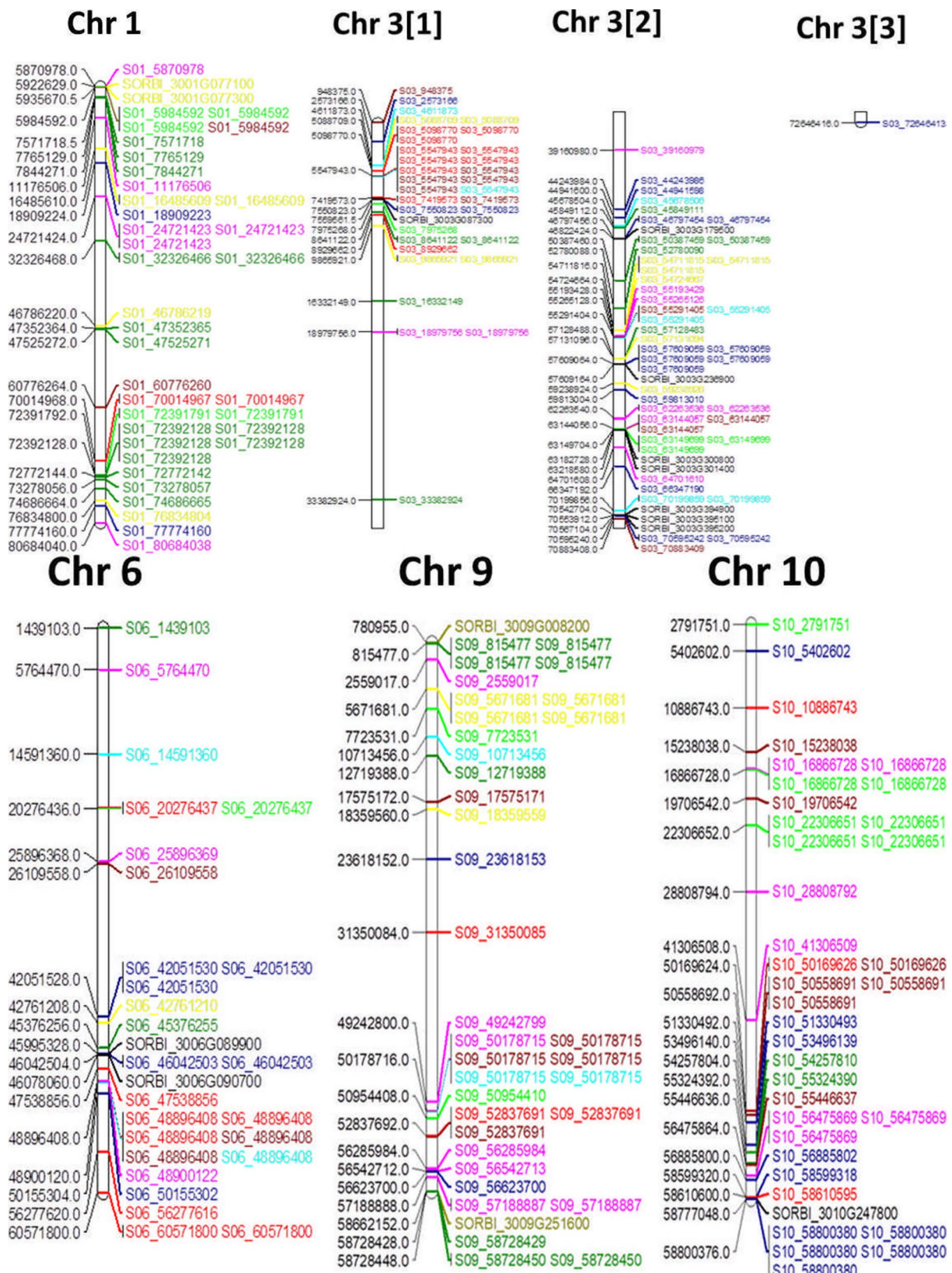
Assessment of the six ML-GWAS methods indicated that pLARMmEB was more significant than the other five models in finding of QTNs. Of the QTNs detected through ML-GWAS, S09\_815477 and S01\_47352365 were found to be significantly associated to RA, which explained 15% and 12% of the phenotypic variation (Table 3). It is obvious that nodal root angle is influenced by multiple genetic factors, environmental conditions, and physiological processes. The identified QTNs explained 15% of the variation, though a substantial

**Table 4** List of colocated QTNs with RA and related trait in the sorghum QTL atlas database

| Trait | QTN          | Chr. | Colocated QTLs               | RA & related traits            | References   |
|-------|--------------|------|------------------------------|--------------------------------|--------------|
| RA    | S01_32326466 | 1    | qRA1_5, qRA1_8               | Nodal root angle               | [24]         |
|       | S01_72392128 | 1    | qRDW1_2                      | Nodal root angle               | [24]         |
|       | S03_8641122  | 3    | QRTAN3.1                     | Root angle of crown root whorl | [60]         |
|       | S05_43703370 | 5    | QRTAN5.1                     | Root angle                     | [24]         |
|       | S07_29563576 | 7    | QRTAN8.1                     | Root angle                     | [24]         |
|       | S07_37035244 | 7    | QRTAN8.1                     | Root angle                     | [24]         |
|       | S09_58728450 | 9    | QRTAN10.1                    | Root angle                     | [24, 39]     |
| RDW   | S03_57609059 | 3    | QRTDW2.1, QRTDW4.2, QRTDW5.1 | Root dry weight                | [24, 31, 61] |
|       | S04_50974937 | 4    | QRTDW4.1, QRTDW8.2           | Root dry weight                | [24, 61]     |
|       | S05_34473660 | 5    | QRTDW8.1                     | Root dry weight                | [61]         |
|       | S06_42051530 | 6    | QRTDW8.1                     | Root dry weight                | [61]         |
|       | S10_58800380 | 10   | QRTDW6.2                     | Root dry weight                | [62]         |
| RL    | S01_24721423 | 1    | QRTL3.3, QRTL5.1, QRTL10.2   | Root length                    | [61–63]      |
|       | S02_6837132  | 2    | QRTL2.1; QRTL5.3             | Root length                    | [61–63]      |
|       | S03_18979756 | 3    | QRTL5.1; QRTL3.3             | Root length                    | [63]         |
|       | S04_1010651  | 4    | QRTL5.3                      | Root length                    | [61, 63]     |
|       | S07_28520448 | 7    | QRTL3.3, QRTL5.1, QRTL10.2   | Root length                    | [62, 63]     |
|       | S08_4855598  | 8    | QRTL5.3, QRTL8.2             | Root length                    | [61]         |
|       | S09_57188887 | 9    | QRTL2.2                      | Root length                    | [64]         |
|       | S10_16866728 | 10   | QRTL3.1                      | Root length                    | [61, 63]     |
|       | S10_56475869 | 10   | QRTL2.2, QRTL2.6             | Root length                    | [61, 63]     |
|       | S01_5984592  | 1    | QRTL2.1, QRTDW8.3            | Root length, root biomass      | [61, 65]     |
| RN    | S01_72391791 | 1    | QRTL3.2, QRTL2.9             | Root length                    | [62, 63]     |
|       | S03_63149699 | 3    | QRTDW4.3                     | Root biomass                   | [65]         |
|       | S04_56437057 | 4    | QRTL5.1                      | Root length                    | [62]         |
|       | S05_16129064 | 5    | QRTL10.1, QRTL10             | Root length, root biomass      | [61]         |
|       | S05_3566674  | 5    | QRTL8.1                      | Root length                    | [63]         |
|       | S05_45567244 | 5    | QRTL10.1                     | Root length                    | [61]         |
|       | S05_67059095 | 5    | QRTL2.7                      | Root length                    | [63]         |
|       | S07_58136512 | 7    | QRTDW2.1                     | Root dry weight                | [61]         |
|       | S10_16866728 | 10   | QRTL10.1, QRTDW8.3           | Root length, root biomass      | [61, 65]     |
|       | S10_22306651 | 10   | QRTL3.1                      | Root length                    | [61]         |

portion of genetic variation remains unexplained. Moreover, the use of ML-GWAS to detect QTNs could potentially help breeders to combine markers to develop marker-assisted selection (MAS) to select plants with improved root angle, speeding up the breeding process [77].

Previous studies on root architectural traits in sorghum have identified multiple Quantitative Trait Nucleotides (QTNs) associated with these traits [24, 30, 35]. For example [24], reported that *qNRA1*, a QTN co-localizing with plant height genes on chromosome 1, and *qNRA5* on chromosome 5, linked to auxin transport genes, influence



**Fig. 11** Chromosomal distribution of significant QTNs and candidate genes for sorghum root architecture traits. **(A)** Linkage groups showing the physical positions of QTNs associated with: RA, green, RDW, yellow, RL, black, PH, red, Rdw\_shDw, blue, RN, orange, ShD, aqua, ShDw, red accent; **(B)** Candidate genes are highlighted as: RA-associated genes (olive green, Chr9), RDW-associated genes (black), RN-associated genes (yellow), Physical distances (bp) between adjacent loci are indicated by scale bars

root angle. Additionally, *qRLN2* (promoting root elongation on chromosome 2) and *qRNO4* (affecting lateral root proliferation on chromosome 4) were identified as key regulators of root length and number [78]. Recent studies have identified pleiotropic QTNs that regulate key drought tolerance traits, including root nodal angle and stay-green properties. For instance, *qDH7* is associated with narrow root angles and delayed leaf senescence, while *qDTy1* influences both root architecture and stay-green duration [72].

In this study, two genes on chromosome 9 were associated with root angle variation (Table S4). The first gene encodes a protein phosphatase 2 C (PP2C), a key regulator of abiotic stress responses, including drought tolerance and ABA (abscisic acid) signaling [80]. The second gene belongs to the Cytochrome P450 (CYP) superfamily, which participates in secondary metabolism and the biosynthesis of defense-related compounds such as phytoalexins and lignins [81]. Additionally, previous studies have identified several genes linked to root architectural traits in sorghum.

This study also identified *ENSRNA049485582*, a gene associated with root number, this gene lies within a known QTL region, and it may encode a regulator of root development and architecture (e.g., auxin-related genes, transcription factors like ARFs, NACs). For instance, in maize, genes such as *PIN1*, *LBD16* have been linked to increased lateral root number, often through their involvement in auxin signaling pathway. Additionally, *SORBI\_3010G247800* linked to root dry weight, was annotated as an E3 ubiquitin-protein ligase *BRE1* [82]. This RING-type E3 ligase is a critical component of the ubiquitin-proteasome pathway, mediating the ubiquitination of histone H2B (at lysine 120 in humans or lysine 123 in yeast) and playing essential roles in transcriptional regulation, chromatin remodeling, and DNA repair. Furthermore, the gene *SORBI\_3003G394900* is part of the *Sobic.003G236900* locus, which encodes a putative disease resistance protein. Also, it was identified as a histone lysine demethylase (KDM) targeting trimethylated lysine 27 on histone H3 (H3K27me3). This enzyme is pivotal in epigenetic gene regulation, particularly in response to abiotic and biotic stresses [83].

This study identified multiple colocalized Quantitative Trait Nucleotides (QTNs) associated with different traits (Table 4), indicating that significant genetic variants for various characteristics overlap within the same genomic regions. Such colocalization offers several advantages for genetic research and crop improvement [84].

In this study, all 33 candidate genes associated with major root architectural traits (RA, Rdw, RL, and RN) were successfully annotated with at least one Gene Ontology (GO) term (Supplementary Table S3, Supplementary Fig. 2).

Functional enrichment analysis using the g: Profiler tool revealed five statistically significant GO terms (Supplementary Table S5), highlighting key biological insights. Particularly, the biological processes included DNA-templated transcription initiation (GO:0006352) and nucleotide excision repair (GO:0006294), suggesting a role for transcription regulation and DNA repair mechanisms in root development.

At the molecular function level, sigma factor activity (GO:0016987) was enriched, implicating regulatory elements in gene expression control. Furthermore, cellular component enrichment identified complexes involved in transcriptional regulation, such as the carboxy-terminal domain protein kinase complex (GO:0032806), nuclear cyclin-dependent protein kinase holoenzyme complex (GO:0000439), and transcription factor TFIID core complex (GO:0005675). Together, these findings point to a coordinated network of transcriptional and DNA repair processes underpinning root architectural traits.

Multi-locus GWAS identified pleiotropic loci associated with multiple root system architectural (RSA) traits highlights the complex genetic regulation underpinning root development in sorghum. Such loci suggest coordinated control of the traits, which are critical for efficient water and nutrient uptake under varying environmental conditions. Understanding pleiotropy could enhance our ability to dissect trait interrelationships. From a marker-assisted selection (MAS) perspective, pleiotropic markers provide a cost-effective avenue to simultaneously improve multiple RSA traits, accelerating the development of ideotypes suited for stress-resilient cropping systems.

## Conclusions

The results revealed significant genetic variation in root angle and related traits within the core collection of Ethiopian sorghum landraces. This serves as a crucial foundation for enhancing these key drought-resilient traits in sorghum. Multi-locus genome-wide association studies for root system architectural traits provide valuable insights for breeding programs aiming to improve root traits through marker-assisted selection, though further validation in diverse environments and functional characterization of candidate genes are necessary to address limitations and guide future research. Ultimately, these findings have the potential to aid in the breeding of sorghum varieties that are more resilient to climate change, emphasizing the valuable genetic resources found in Ethiopian sorghum landraces for developing water-efficient cultivars.

## Abbreviations

|      |                                 |
|------|---------------------------------|
| RSA  | Root system architecture        |
| QTN  | Quantitative trait nucleotides  |
| SNPs | Single nucleotide polymorphisms |

|         |                                           |
|---------|-------------------------------------------|
| ML-GWAS | Multi-locus genome wide association study |
| LOD     | Logarithm of odds                         |
| MAF     | Minor allele frequencies                  |
| GBS     | Genotyping by sequencing                  |
| LD      | Linkage disequilibrium                    |
| QQ      | Quantile quantile                         |
| PCA     | Principal component analysis              |
| RA      | Root angle                                |

## Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12870-025-07271-6>.

Supplementary Material 1

## Acknowledgements

The author would like to thank the National Sorghum and Millet Improvement Program at Melkassa Agricultural Research Center, Ethiopia, for supplying the experimental planting materials.

## Author contributions

Abel Debebe Mitiku: conceptualization, methodology, data analysis, writing-original draft; Tileye Feyissa: conceptualization, methodology, supervision, writing-review and editing; Alemu Tirfessa Woldetensaye: conceptualization, methodology, supervision, writing-review and editing; Habte Nida Chikssa: conceptualization, methodology, writing-review and editing; Temesgen Mawos Menamo: conceptualization, methodology, writing-review and editing; Tewodros Mesfin Abebe: conceptualization, methodology, writing-review and editing; Kassahun Bante: conceptualization, methodology, writing-review and editing.

## Funding

Financial support of the experiments was provided by Ethiopian Institute of Agricultural Research and the Food System Resilience Project Addis Ababa, Ethiopia.

## Data availability

The original contributions presented in the study are included in the article; the source link for the SNP data: <https://doi.org/10.1002/tpg2.20055>. Further inquiries can be directed to the corresponding author.

## Declarations

### Ethics approval and consent to participate

Not applicable. Clinical trial number: Not applicable.

### Consent for publication

Not applicable.

### Competing interests

The authors declare no competing interests.

Received: 23 May 2025 / Accepted: 19 August 2025

Published online: 02 September 2025

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